

HOLY ANGEL UNIVERSITY

SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS

A Capstone Project
Presented to the Faculty of the
School of Computing
Holy Angel University



In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology with Area of Specialization in Web
Development

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*Proposal manuscript — draft (Introduction to Conceptual Framework). Later sections follow in the complete proposal.
Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are
placeholders.*

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APPROVAL SHEET

This capstone entitled “**SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS**”, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development has been examined and is recommended for acceptance and oral examination.

Prof. Charlie
 Adviser

ORAL EXAMINATION

Approved by the committee of oral examiners on
 [Date of defense]

Prof. Alpha
 Panel Chair

AD Echo
 Panel Member

[Name of Panel Member]
 Panel Member

APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development.

Prof. Alpha
 Program Chair/Coordinator,
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Dean Delta
 Dean, School of Computing

HOLY ANGEL UNIVERSITY**Acknowledgment**

The researchers extend their sincere gratitude to everyone who supported this capstone project. This page is placeholder text: a real manuscript thanks the Adviser, the panel, the instructors, the host office and the families who made the work possible.

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Table of Contents

	Page
Cover Page	i
Approval Sheet	ii
Acknowledgment	iii
Table of Contents	iv
Title Page	1
Introduction	2
Review of Related Literature	2
Queueing Theory and Perceived Waiting Time	2
Web-Based and Mobile Queue Management Systems	3
Notification Channels and Response Behavior	3
Software Quality Evaluation	3
Synthesis	3
Conceptual Framework	3
Objectives of the Study	4

HOLY ANGEL UNIVERSITY**Smart Queue Management System for University Clinics**

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Abstract

The abstract is written after the study is completed and is left blank at the proposal stage.

HOLY ANGEL UNIVERSITY**SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS****Introduction**

University clinics serve a steady stream of students, faculty and staff who arrive for consultations, medical clearances, dental check-ups and the release of laboratory results. At most campuses the arrival of these patients is still managed by a paper logbook at the reception desk and a verbal call of names once a nurse or physician is free.

Digital queue management systems address this by issuing a numbered ticket, displaying the current number, and notifying the patient when their turn approaches. Commercial systems are priced and designed for large institutions; a lightweight web-based system running on the phones students already carry would suit a campus clinic better.

The problem this study addresses is the following: patients cannot see their position in line, staff are interrupted by repeated questions, urgent cases wait behind routine clearances, and the clinic cannot report service volume. This capstone project therefore proposes the Smart Queue system.

Review of Related Literature

This section reviews studies and existing systems related to the Smart Queue system and the methods used to evaluate it.

Queueing Theory and Perceived Waiting Time

For a single-server clinic desk, the expected waiting time grows sharply as utilization approaches one (Example & Writer, 2022). Uncertain waits feel longer than known, finite waits of the same duration (Placeholder, 2021), which supports giving every patient a live view of the queue.

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Web-Based and Mobile Queue Management Systems

A registrar queue built as a progressive web application reduced counter congestion by letting students wait outside and return when notified (Author et al., 2023). The proposed system adopts ticket issuance without a kiosk, a public display and staff call controls.

Notification Channels and Response Behavior

Email and push notifications are nearly free to send, while SMS carries per-message costs small institutions struggle to sustain (Writer, 2022).

Software Quality Evaluation

Capstone projects in the Philippines commonly ask IT experts to rate a system against ISO/IEC 25010 characteristics on a Likert scale (Example, 2023).

Synthesis

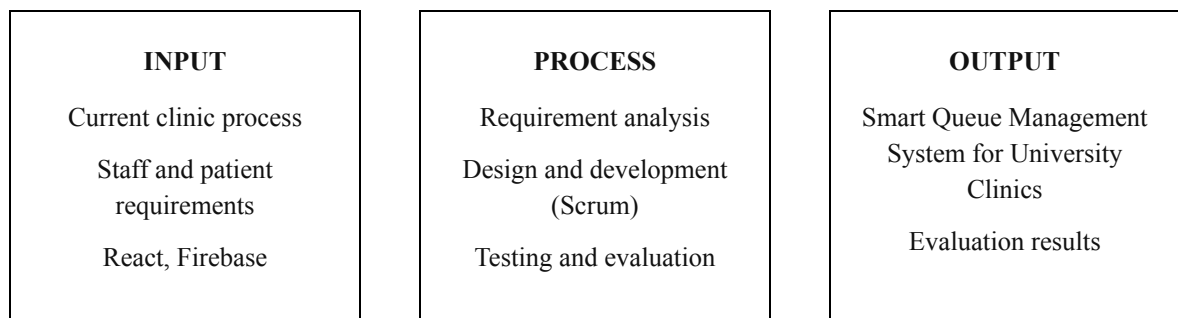
The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the Smart Queue system fills.

Conceptual Framework

The study follows the Input–Process–Output model shown in Figure 1.

Figure 1

Conceptual Framework of the Study



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Draft: Objectives of the Study, Scope and Delimitations and Method are being written and will be in the complete proposal manuscript.